KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 8 – Practice Examination 2

Equations, Investigating Data and Mixed Topics

Practice paper for the Year 8 Yearly Examination

Instructions		
1. Attempt all questions 2. Show working where needed 3. Time Allowed 60 minutes 4. Calculators are permitted 5. Marks are shown beside each question	Equations (Section A):	/18
	Investigating Data (Section B):	/21
	Mixed Topics (Section C):	/21
	Total Marks:	/60

Section A – Equations (18 marks)

Suggested time: 20 minutes. Show all working.

Question 1 – One-step equations **(4 marks)**

Solve each equation.

a) $x + 13 = 30$ x = _____ **(1 mark)**

Working: _____

b) $m - 8 = 19$ m = _____ **(1 mark)**

Working: _____

c) $9p = 63$ p = _____ **(1 mark)**

Working: _____

d) $y \div 6 = 8$ y = _____ **(1 mark)**

Working: _____

Question 2 – Two-step equations **(4 marks)**

Solve each equation, showing your working.

a) $6x - 4 = 32$ **(2 marks)**

Working: _____

_____

_____

x = _____

b) $x \div 5 + 7 = 11$ **(2 marks)**

Working: _____

_____

_____

x = _____

Question 3 – Equations with a negative solution **(2 marks)**

Solve $4m + 25 = 5$.

Working: _____

_____

_____

m = _____

Question 4 – Checking solutions **(2 marks)**

Use substitution to show whether $x = 5$ is the solution to $2x + 11 = 21$.

Working: _____

_____

_____

_____

Is x = 5 the solution? (circle one) YES / NO

Question 5 – Worded problem **(3 marks)**

I think of a number, multiply it by 5, and then subtract 9. The answer is 41.

a) Write an equation to represent this problem. Use *n* for the number. **(1 mark)**

Equation: _____

b) Solve your equation to find the number. **(2 marks)**

Working: _____

_____

_____

n = _____

Question 6 – Worded problem **(3 marks)**

Ava hires a bike. She pays $8 to hire a helmet plus $6 for every hour she has the bike. She pays $38 altogether. Let *h* be the number of hours she hires the bike.

a) Write an equation to represent this situation. **(1 mark)**

Equation: _____

b) Solve your equation to find how many hours Ava hired the bike. **(2 marks)**

Working: _____

_____

_____

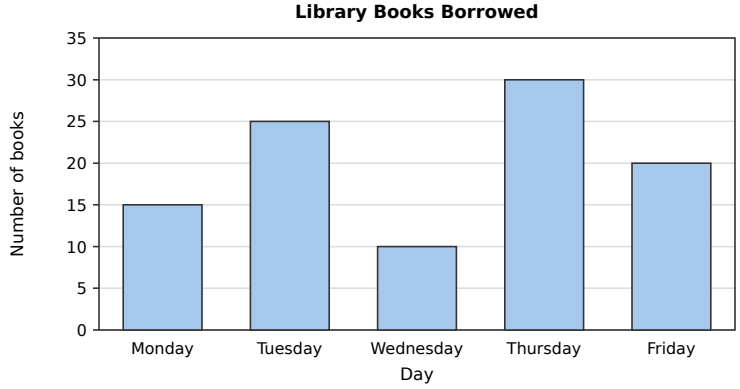
h = _____

Section B – Investigating Data (21 marks)

Suggested time: 20 minutes. Show all working.

Question 7 – Column graph **(2 marks)**

The column graph shows the number of books borrowed from the school library each day for one week.



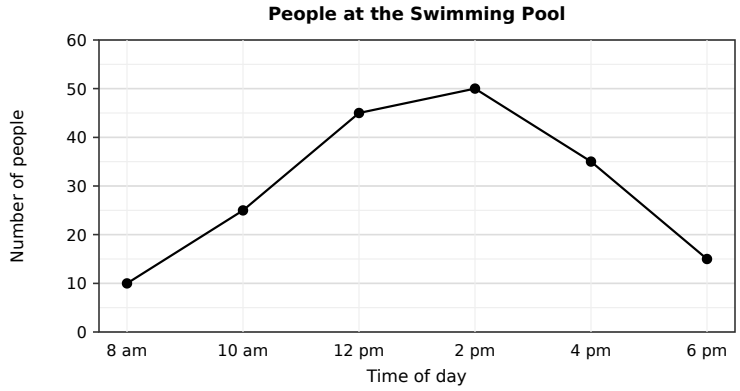
a) On which day were the most books borrowed? _____ **(1 mark)**

b) How many books were borrowed altogether during the week? _____ **(1 mark)**

Working: _____

Question 8 – Line graph **(2 marks)**

The line graph shows the number of people at a swimming pool at different times of the day.



a) At what time were there the most people at the pool? _____ **(1 mark)**

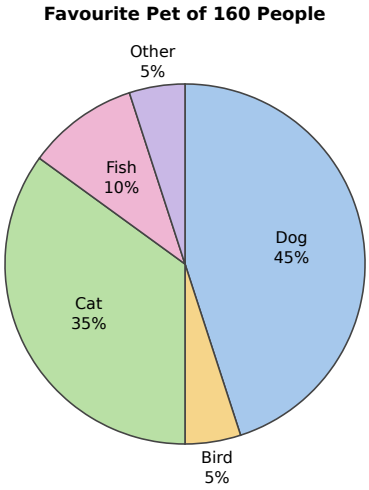
b) By how many people did the number at the pool decrease between 2 pm and 6 pm? **(1 mark)**

Working: _____

Answer: _____

Question 9 – Sector (pie) graph **(3 marks)**

160 people were asked what type of pet they would most like to own. The results are shown in the sector graph.



a) How many people chose a cat? **(1 mark)**

Working: _____

Answer: _____

b) How many more people chose a dog than a bird? **(2 marks)**

Working: _____

_____

_____

Answer: _____

Question 10 – Stem-and-leaf plot **(2 marks)**

The ages of 13 people at a family party are shown in the stem-and-leaf plot.

Stem	Leaf
1	5 8 9
2	2 4 4 6 8
3	0 3 7
4	1 6

Key: 3 | 0 means 30

a) Find the median. _____ **(1 mark)**

Working: _____

b) How many people were older than 30? _____ **(1 mark)**

Question 11 – Dot plot **(2 marks)**

A soccer team recorded the number of goals it scored in each of its 18 games. The results are shown in the dot plot.



a) What is the mode? _____ **(1 mark)**

b) In how many games did the team score at least 3 goals? _____ **(1 mark)**

Question 12 – Frequency table **(2 marks)**

The number of people in each car that dropped students at school one morning is shown in the frequency table.

People in the car	1	2	3	4	5
Frequency	12	15	6	4	3

a) How many cars were surveyed altogether? _____ **(1 mark)**

Working: _____

b) What is the mode? _____ **(1 mark)**

Question 13 – Mean, median, mode and range **(5 marks)**

The points scored by a basketball player in six games are shown below.

18, 24, 15, 24, 30, 21

a) Calculate the mean. **(2 marks)**

Working: _____

_____

_____

Mean = _____

b) Find the median. _____ **(1 mark)**

Working: _____

c) Find the mode. _____ **(1 mark)**

d) Find the range. _____ **(1 mark)**

Working: _____

Question 14 – Outliers **(3 marks)**

The heights, in centimetres, of six seedlings are shown below.

23, 25, 22, 4, 24, 28

a) Identify the outlier in this data set. _____ **(1 mark)**

b) Calculate the mean of the data, including the outlier. **(1 mark)**

Working: _____

_____

Mean = _____

c) Does the outlier make the mean higher or lower than most of the data values? Explain your answer. **(1 mark)**

Working: _____

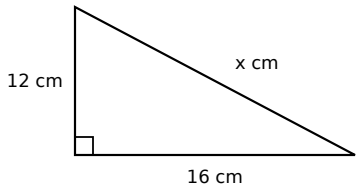
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Section C – Mixed Topics (21 marks)

Suggested time: 20 minutes. Show all working.

Question 15 – Pythagoras' Theorem **(4 marks)**

a) Find the length of the hypotenuse, *x*, in the triangle below. **(2 marks)**



Working: _____

_____

_____

x = _____

b) A triangle has side lengths 6 cm, 9 cm and 11 cm. Is it a right-angled triangle? Show working. **(2 marks)**

Working: _____

_____

_____

_____

Question 16 – Decimals **(2 marks)**

a) Calculate 2.6 × 0.15 _____ **(1 mark)**

Working: _____

_____

b) Calculate 9.6 ÷ 0.08 _____ **(1 mark)**

Working: _____

_____

Question 17 – Powers and roots **(2 marks)**

a) Evaluate 2^5 _____ **(1 mark)**

Working: _____

b) Evaluate √144 _____ **(1 mark)**

Working: _____

Question 18 – Surds **(3 marks)**

a) Circle the surds: √20 √49 √30 √100 **(1 mark)**

b) Simplify √72 **(2 marks)**

Working: _____

_____

_____

Answer: _____

Question 19 – Algebra **(3 marks)**

a) Simplify $7m - 2n + 3m + 6n$ _____ **(1 mark)**

Working: _____

b) Expand $4(y - 5)$ _____ **(1 mark)**

Working: _____

c) Evaluate $3a^2 - b$ when $a = 3$ and $b = 7$. _____ **(1 mark)**

Working: _____

_____

Question 20 – Area and volume **(3 marks)**

a) Find the area of a trapezium with parallel sides 9 cm and 13 cm and height 6 cm. **(2 marks)**

Formula: Area = ½ × (a + b) × h, where a and b are the parallel sides and h is the perpendicular height.

Working: _____

_____

_____

Answer: _____

b) Find the volume of a rectangular prism that is 12 cm long, 3 cm wide and 5 cm high. **(1 mark)**

Working: _____

_____

Answer: _____

Question 21 – Linear relationships **(2 marks)**

Complete the table of values for the rule $y = 4x - 1$.

x	−2	−1	0	1
y				

Question 22 – Rates and ratios **(2 marks)**

a) Write the ratio 20 : 32 in simplest form. _____ **(1 mark)**

Working: _____

_____

b) A train travels 450 km in 5 hours. Find its average speed in km/h. **(1 mark)**

Working: _____

_____

Answer: _____

END OF EXAMINATION PAPER